CISCO–AICTE VIRTUAL INTERNSHIP 2026

NetSage AI

An AI-Assisted Dashboard for Network Fault Diagnosis

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Track: Applied AI + Network Troubleshooting

What This Covers

01 **The Problem**

Why junior engineers struggle to trace symptoms to root cause

02 **What I Built**

A demo-ready dashboard — and where it differs from the original plan

03 **Features**

Monitoring Overview, Case Explorer, Responsible AI Log

04 **Honest Scorecard**

Planned deliverables vs. what actually shipped

05 **Next Steps**

What I'd build next, in priority order

The Problem

“

A PC gets an IP address but still can’t reach the server.

A symptom every junior engineer recognizes — but the root cause could be hiding almost anywhere.

VLAN

Routing

DHCP

DNS

ACL

NAT

Wireless

Any one of these layers could be the real root cause — tracing it takes experience a junior engineer hasn’t built up yet.

What I Actually Built

A single-page monitoring dashboard, built with Lovable, running on mock data — designed to be screen-recorded as a demo.

ORIGINAL PLAN

- ✦ Live Claude API call for diagnosis
- ✦ CSV import pipeline for cases
- ✦ Separate deterministic rule-checker
- ✦ 30+ case dataset
- ✦ Full dashboard + audit log

WHAT SHIPPED

- ✓ Pre-written mock diagnoses (no live API)
- ✓ 12 hardcoded cases in mock-cases.ts
- ✓ No rule-checker module yet
- ✓ 8 fault categories represented
- ✓ Dashboard + Case Explorer + RAI Log — all working

Tech Stack

1

React 18 + TypeScript

Core application framework

2

TanStack Start / Router

Routing and app structure

3

Vite

Build tooling and dev server

4

Tailwind CSS

Styling system

5

shadcn/ui

Component library

6

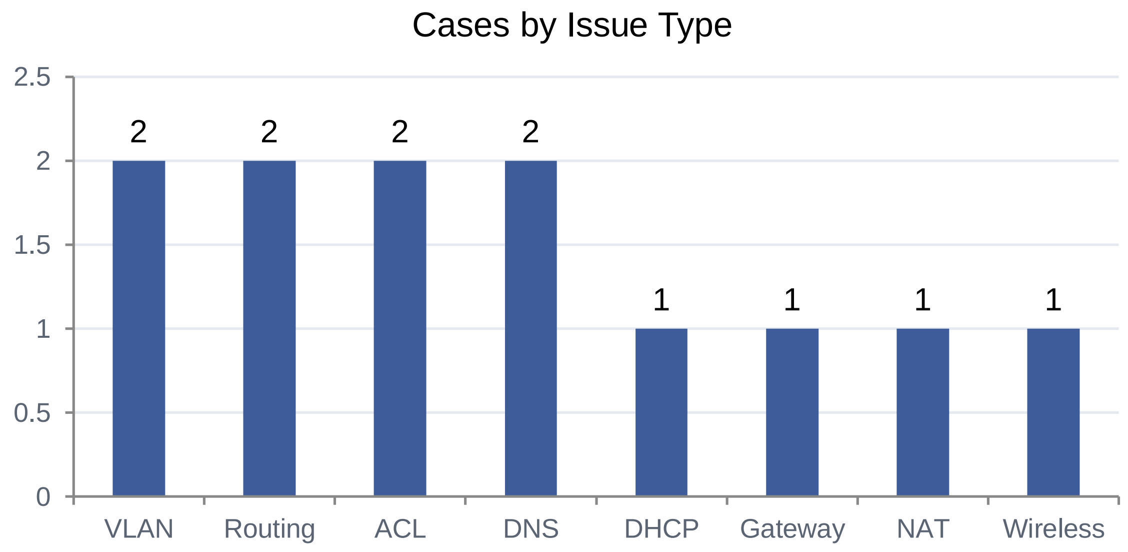
Recharts

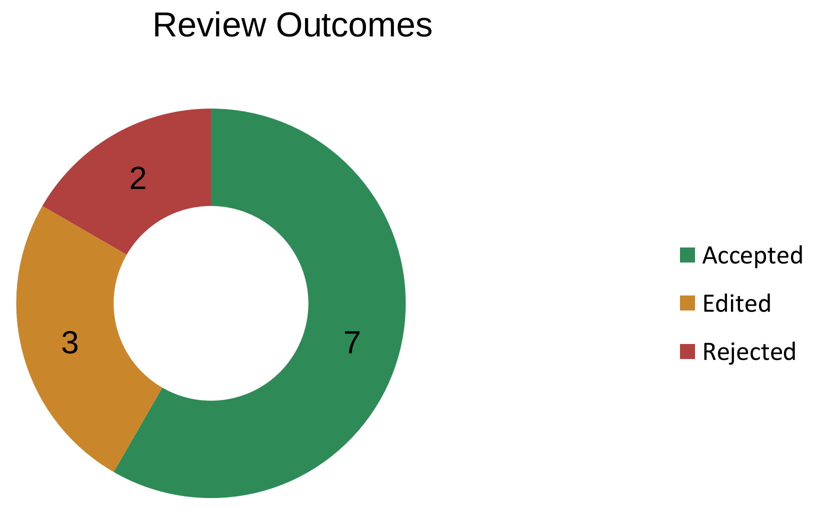
Bar chart and donut chart visualizations

Feature: Monitoring Overview

The dashboard landing tab — a snapshot of case volume and how often the AI and the human reviewer agreed.

12	8	3
Total Cases	Fault Categories	Review Outcomes Tracked





Feature: Case Explorer

Every case, filterable and sortable — click through to see the AI diagnosis next to the reviewer's verdict.

Symptom	Fault Type	Confidence	Status
PC gets IP, can't reach server	DHCP scope mismatch	92%	Accepted
Intermittent packet loss on WLAN	Wireless channel overlap	78%	Edited
Two VLANs can't inter-communicate	Missing SVI route	88%	Accepted
Website loads by IP, not by name	DNS forwarder misconfigured	65%	Rejected

Row click opens a detail panel: root cause → evidence → next command → fix steps, alongside the reviewer's note.

Feature: Responsible AI Log

The human-in-the-loop record — every time a reviewer edited or rejected the AI's answer, and why.

Edited

Wireless channel overlap

AI missed the co-channel interference from a neighboring AP — reviewer added it to the fix steps.

Rejected

DNS forwarder misconfigured

AI's root cause was plausible but the actual issue was a stale ACL blocking port 53 — reviewer corrected it.

Why it matters: showing the AI's answer isn't enough — you need a record of when and why a human overruled it.

Honest Scorecard

Planned Deliverable	Status
Case dataset (30+ cases)	12 built
AI diagnosis output	Mocked
Dashboard (overview + charts)	Done
Responsible AI log	Done
Rule-based checker	Not started
Demo video	Pending

What I'd Build Next

1

Wire up the live Claude API

Replace the mock fields with a real model call for diagnosis

2

Add the rule-checker module

An independent, deterministic cross-check against the AI's answer

3

Push past 30 cases

Via CSV import rather than more hand-typed entries

4

Record the walkthrough demo

Screen-record the review flow and share via Google Drive

Certifications Completed

Three Cisco Networking Academy courses, completed through IILM University alongside the project.

✓

Data Analytics Essentials

11 Jun 2026

✓

Introduction to Modern AI

11 Jun 2026

✓

Apply AI: Analyze Customer Reviews

11 Jun 2026

Thank You

The workflow is proven; the live API, rule-checker, and larger case set are the clear next steps.

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